



ST BENEDICT'S COLLEGE

SOURCE BOOKLET
LIFE SCIENCES
NOVEMBER EXAMINATION
GRADE 11

QUESTION 1

Read the information on the establishment of a game reserve in KwaZulu-Natal and then answer the questions that follow.

Colchester Zoo is an innovative and world-renowned zoo in England. It focuses on conservation of endangered species and captive breeding programmes. In 2005 it decided to set up its own conservation project and chose South Africa to create a private nature reserve.

Creating the UmPhafa Private Nature Reserve

An area of farmland three hours north of Durban was deemed a suitable location to set up the UmPhafa Private Nature Reserve. This 5,000 hectare area comprised of three farms previously farmed for cattle, with a wide range of habitats from acacia, savannah and bushveld to mountainous terrain. The reserve is situated along the banks of the Tugela River. Owing to previous agricultural practices, many of the natural inhabiting species had been lost and much of the land had been eroded because of overgrazing. Historically this part of KwaZulu-Natal has been an important agricultural area. There are many farms that are under cultivation with crops such as maize and soya. Many types of livestock are farmed here, e.g. chickens, pigs, beef and dairy farms, which flourish in this part of South Africa. The aim of Colchester Zoo management was to rehabilitate UmPhafa and re-establish the region's previous natural habitat. UmPhafa is a private nature reserve and is not open to tourists. It serves mainly as a conservation and research facility.

Figure 1: Releasing Impala onto UmPhafa Private Nature Reserve



Fencing and animal release

The first step in the management of the reserve was to remove all of the old cattle fencing and equipment and then to start the immense task of fencing the entire reserve. Once the fence around the farm was electrified, an animal release programme started. Many of the released species are increasing naturally, but in order to ensure adequate habitat management, the numbers of grazers will need to be increased. By introducing more individuals, it will prevent over-domination of certain grass species and promote floral diversity. These released species will also eventually serve as a prey base for larger predators.

Mammal species

Some mammal population remained on UmPhafa and have thrived since the reserve was established. However, since 2006, many more species have been reintroduced. The reserve is now home to a wide range of fauna, including White Rhino and giraffe. Many buck species have also been brought in, such as kudu, impala, reedbuck, zebra, blesbok, red hartebeest, blue wildebeest, waterbuck, and nyala. Warthog, aardvark and porcupine have also been brought in. Small to medium-sized carnivores, such as black-back jackal, caracal, serval, genet, and several species of mongoose are also common and recently the staff have been noticing more and more signs of leopard.

Breeding success

Since release, numerous species have bred on the reserve, including all of the antelope species. In addition, nine giraffe and two white rhino calves have been born. Eventually these species will reach carrying capacity on the farm. Signs of leopard are increasing, due to an increase in prey items, but the eventual aim is to release other carnivorous species, such as cheetah, onto the reserve to ensure prey population are managed sustainably.

Figure 2: Nine giraffe have been born on the reserve since they were reintroduced



Goals of UmPhafa

To date, all funding for UmPhafa has been provided by Colchester Zoo and donations from its visitors. The goal, however, is that UmPhafa becomes self-sufficient. Another priority for UmPhafa is to work with the local community. At present, thirteen members of the local community are employed in positions ranging from game guards to fencing staff. All staff are valued and their skill enhanced. To date, the game guards have been put on game ranger and wildlife management courses to increase their knowledge. Coupled with employment, UmPhafa staff has also been working with local schools. An education programme has commenced at schools to incorporate wildlife species into their curriculum. Fundraising has also taken place for these schools, with computers purchased for one school and a new kitchen erected at the other; purchased with funds generated by two schools in England. The eventual aim is to raise sufficient funds to construct classroom facilities on UmPhafa so that the children can visit the reserve for environmental lessons. Coupled with the educational programme and recent meetings concerning a recycling centre for the local community on site, UmPhafa is well on its way to becoming a creditable local conservation project.

Figure 3: Population parameters

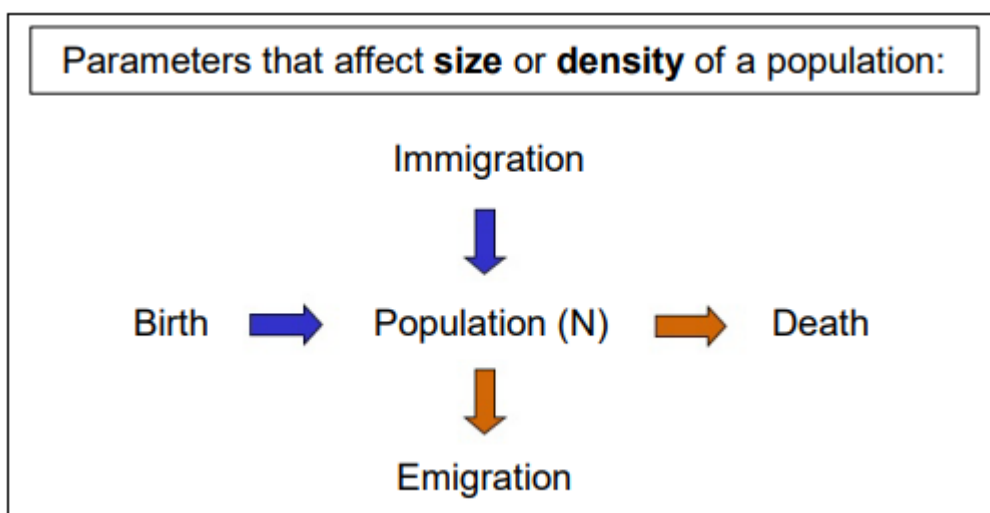
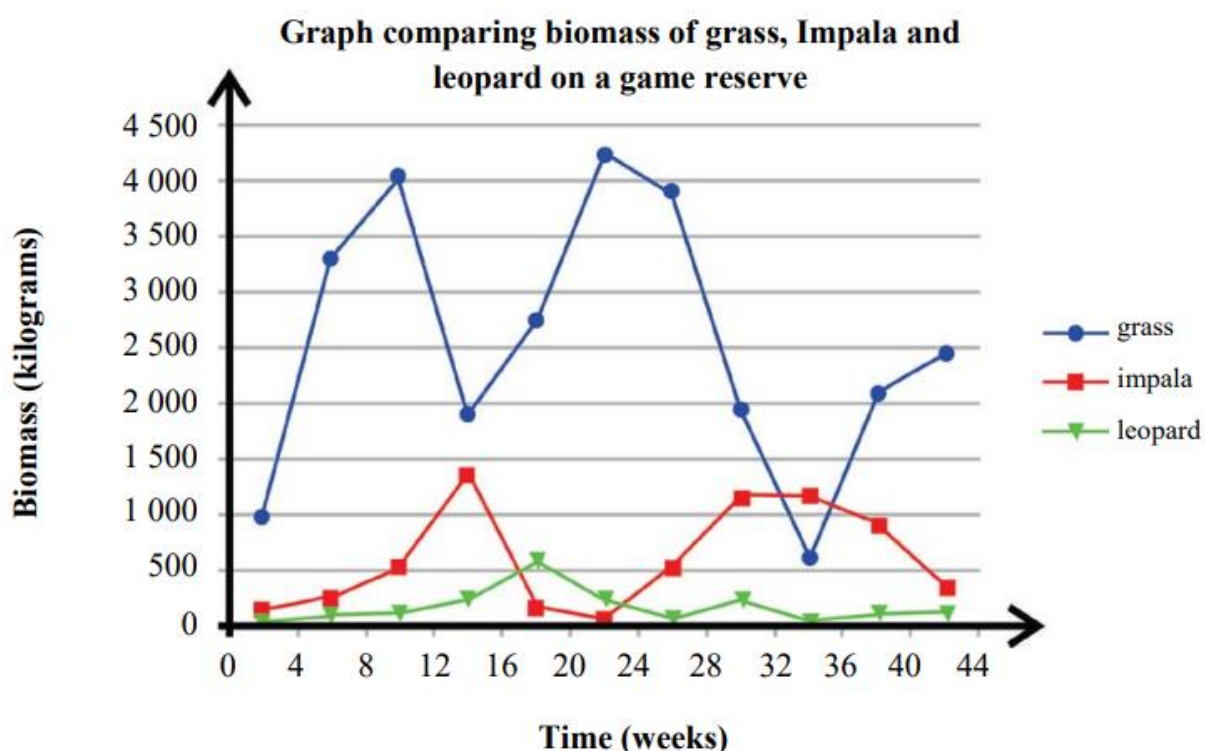


Figure 4: Possible relationship between the biomass of the Leopard, Impala and Grass on UmPhafa



QUESTION 2

Read the information below on the zebra population of the Makgadikgadi Pans National Park in Botswana

Nothing Can Stop the Zebra – A 150-mile fence in the Kalahari Desert appeared to threaten Africa's zebras, but now researchers can breathe a sigh of relief.

Zebra migrating to the Makgadikgadi Pans



[Source: <<http://smithsonianmag.com>>]

Zebras undertake migrations every year in search of water and grazing. They can cover up to 480 kilometres a year as they travel and the migration involves up to several thousand zebras. The results of studies on their migrations were astounding to researchers when they realised that zebras, large and highly visible animals, moved this kind of distance and appear to do so every year. Zebra migration routes in many parts of Africa have been disrupted, often by fence construction. As a result, zebra populations diminish and local economies that depend on ecotourism may suffer. Phenomena like this newly discovered migration show these animals need large, wide-open spaces in order to survive.

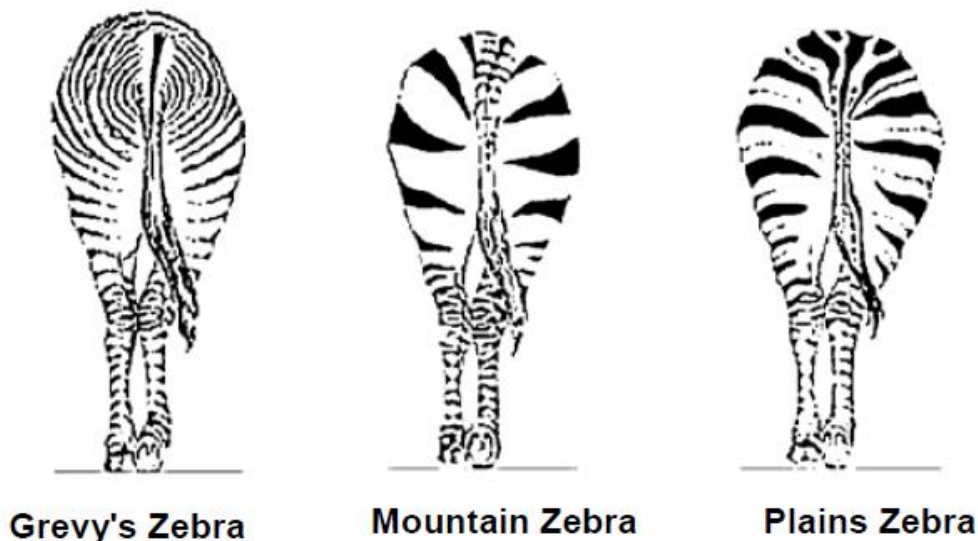
The fence, which the Botswana government installed on the western edge of the Makgadikgadi Pans National Park in 2004, was primarily intended to protect cattle on neighbouring farmlands from lions that live in the park. But wildlife experts became concerned about the barrier's impact on zebras. They had reason to worry, given Botswana's disastrous history with fences. In 1983, during a devastating drought, at least 50,000 wildebeest died because a fence blocked their route to water.

But trying to predict how the new fence would affect the vast zebra herds that rely on that expanse of land for grazing and water was no easy task. Christopher Brooks, a scientist who works on conservation projects in Africa, started the Zebra Research Project as he was "concerned that a fence could have serious negative consequences, but there was no solid ecological data" about the zebras and their migration. Despite being among the most recognisable of large African animals, zebras and their extraordinary movements turn out to be rather mysterious.

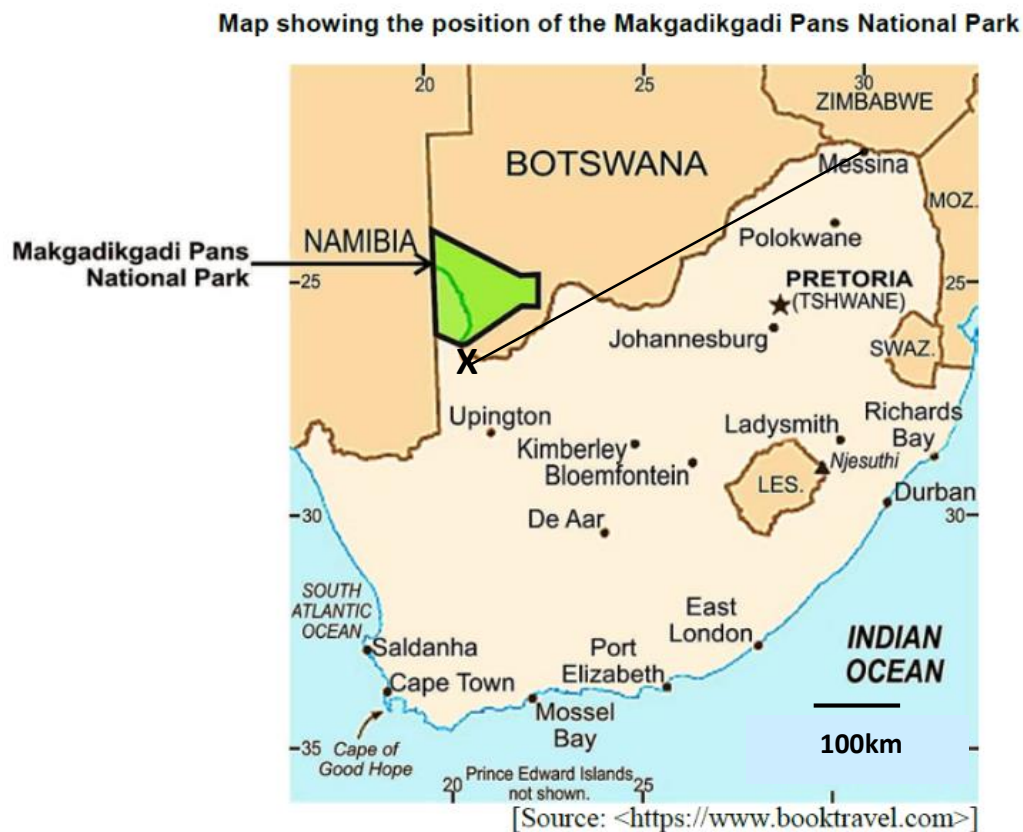
Jonathan Bradley, a biologist from Bristol University, has taken over the running of the Makgadikgadi Zebra Migration Research Project, which is attempting to answer the question: Would a two-metre high electrified fence stretching 150 miles across the zebras' territory disrupt their migration? The project aims to understand the impact of fencing policies on all wildlife across Africa.

Zebras come in three distinct species: Plains, Mountain and Grevy's; Plains zebra are the most widespread, occurring throughout much of southern and eastern Africa. As members of the *Equus* genus, they are closely related to horses. Zebras are not well suited to domestication since they are unpredictable and have been known to attack people trying to handle them. Like a human fingerprint, a zebra's strip pattern is unique. There are many theories about why the stripes evolved. The dizzying lines might distort a zebra's outline, for instance, or make the animal look bigger, confusing predators.

Hindquarters of three zebra species showing differences in stripes



During the dry season, zebras live along the Boteti River, the only regular source of water. When the rains come, in early summer, the herds move to open grassland, where temporary pools fill with water, and then on to the rain-filled salt pans, where nutritious grasses grow on the edges. Zebras are regarded as a keystone species in the Makgadikgadi. Leading the migration, zebras eat longer grasses, exposing short, sweet shoots for the more selective wildebeest that trail them, while the small population of springbok, bringing up the rear, must settle for leftovers. Then there are the predators that zebras sustain. Lions eat them and brown hyenas scavenge their carcasses.



Bradley has fitted radio collars to ten zebra mares providing him with valuable insights into zebra migrations within southern Africa. He tracks zebras and maps out their movements. The mares are often from different social units. A social unit may consist of 50 or so zebra made up of a lone stallion, one to six mares and all their offspring. These small, tightknit families come together by the thousands for the seasonal migration in search of grass and water.

Stallions fight to protect mares in their harems or abduct mares in heat. (Bradley tracks mares rather than stallions because the females are less likely to fight with each other and damage the collars.) The ties that bind a stallion and his harem are profound. Bradley once noticed a lone stallion standing for hours in the riverbed, not eating. When Bradley approached, he saw that the stallion was standing vigil over a dead mare.

This young zoologist has witnessed this single-minded devotion when he has darted mares to collar them. "Once the tranquilisers start to take effect, some stallions bite on the females' necks to try to keep them upright and moving," he says. "While we're busy with the female, the stallion moves through the herd, constantly calling, looking for his missing mare. When she wakes up and calls, the stallion heads directly to her." Mares, too, are loyal, often remaining with a single harem for life, a period that can span 16 years.

By tracking the zebras' movements, Brooks and Bradley discovered that zebras are more resilient than previously thought. The Makgadikgadi researchers recorded them trekking in dry months more than 34 kilometres a day to preferred grazing lands. During such trips, the animals go without water for up to seven days.

In 1989, after years of drought, the Boteti River dried up. Too many animals were competing for too little water. Elephants bullied zebra and wildebeest. Prowling lions set off terrifying stampedes of zebras. In 2007 in an attempt to protect wildlife during the drought, government authorities and lodge owners dug holes and filled them with water. "The zebra stood 20 yards away, watching us dig. When we pumped the first water, they were there in an instant," says Bernie Esterhuyse from Leroo La Tau safari lodge. "I had tears in my eyes when I saw them finally drink in peace."

Ten years into the Zebra Migration Project, Bradley and his colleagues can report the species is thriving. Early indications are that the Makgadikgadi fence does not restrict their migration. Reduced competition from cattle has meant more grazing for zebras inside the park. More zebra foals are surviving beyond their first year, and the population appears to be growing.

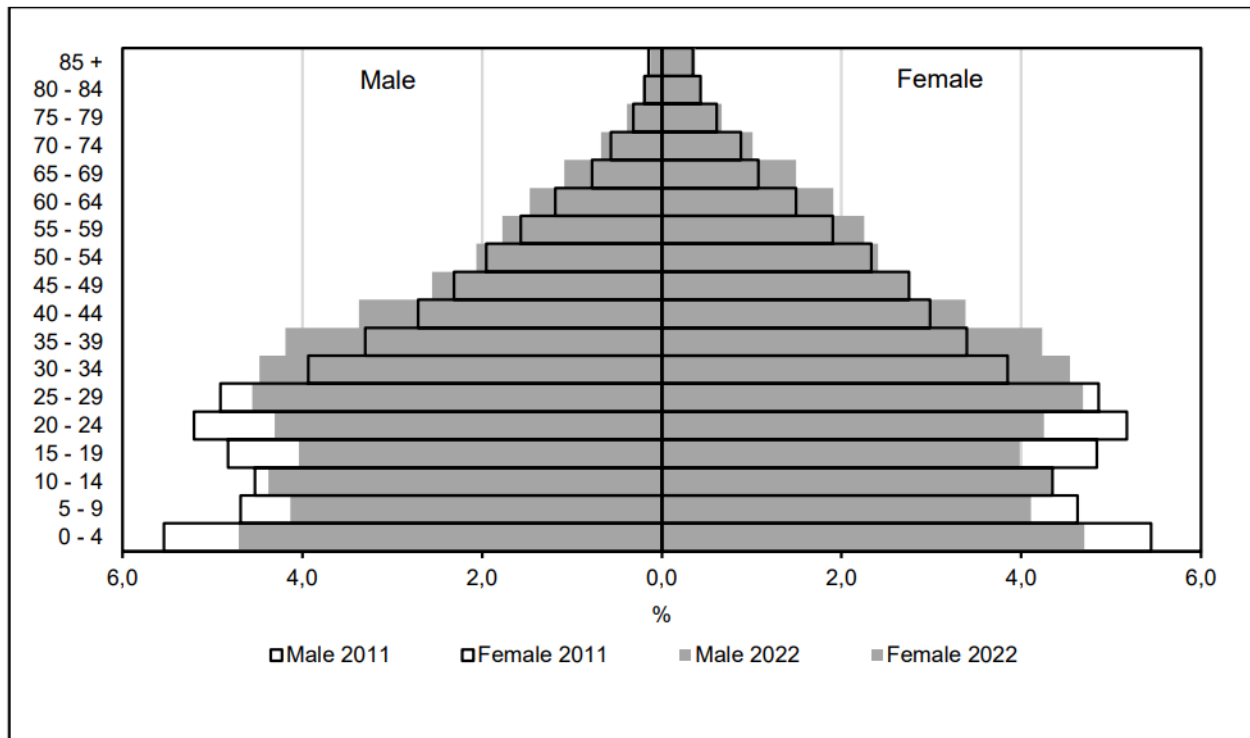
[Adapted from: <<http://www.smithsonianmag.com>> March 2011]

QUESTION 3

SOURCE A Population Statistics 2011 to 2022

Percentage distribution of the total population of South Africa by age group and sex

Figure 3.4: Distribution of the population by age group and sex, Census 2011 (transparent) and Census 2022 (shaded)



Adapted from: www.statssa.gov.za

SOURCE B World Population Prospects – Global Demographic Estimates and Projections by the United Nations

The last national census of the South African population was carried out in 2022.

South Africa Population History

At the 2011 Census, the final declared total for resident citizens in South Africa was 51 770 560. In terms of population density that equated to 43 people living in every square kilometre of land and the country's sparseness was highlighted by the fact that it was only the 169th biggest in the world for density alone.

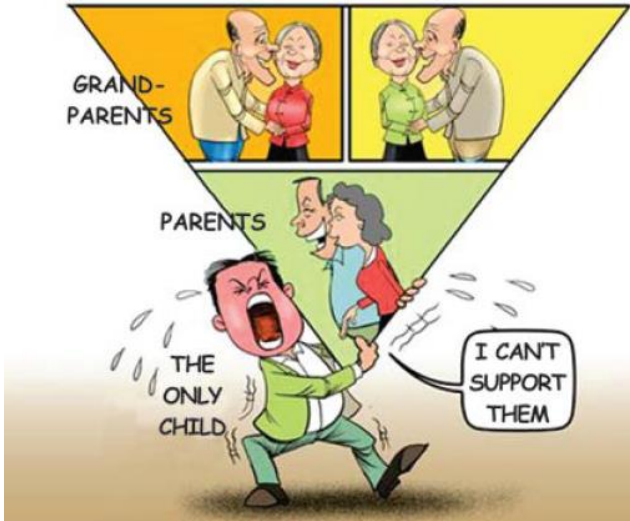
The finding in 2022 suggest that there had been a significant rise in numbers over the course of the previous 21 years and that the South African population has grown to 62 024 504, making it the 24th largest country in the world.

Table showing population statistics from 2015 to 2024

YEAR ~	POPULATION	CHANGE	DENSITY (/KM ²)	POPULATION RANK	DENSITY RANK
2024	64,007,200	1.44%	53	24	160
2023	63,212,400	1.52%	52	24	160
2020	60,562,400	1.64%	50	24	160
2019	59,587,900	1.66%	49	24	160
2018	58,613,000	1.7%	48	24	160
2017	57,635,200	0.66%	48	24	160
2015	56,723,500	1.62%	47	24	160

[Adapted from: <<http://worldpopulationreview.com>>]

SOURCE C Labour Force



Age dependency ratio

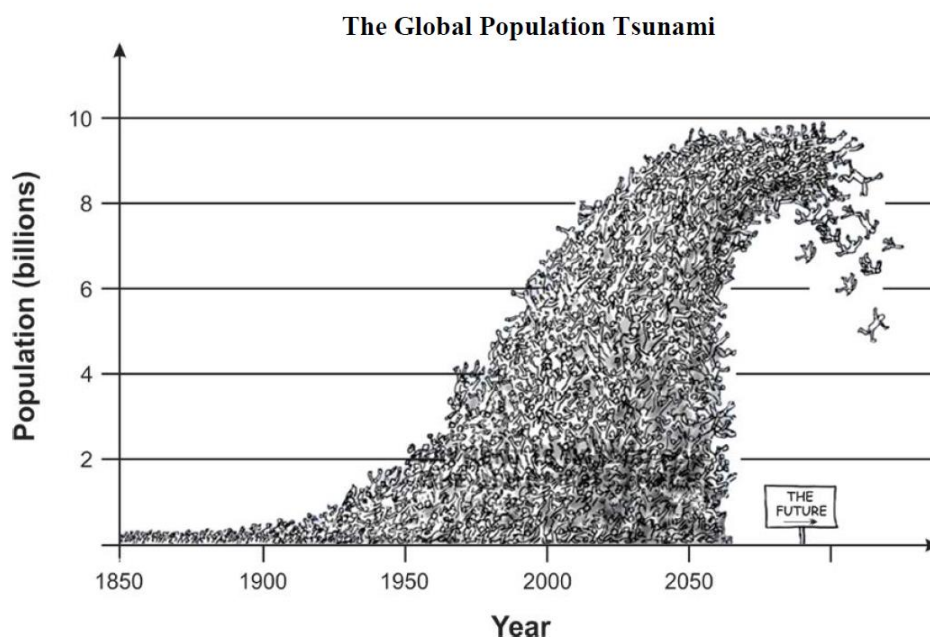
The dependency ratio of a population is a ratio of people who are generally not in the labour force (the dependents) to the workforce of a country (the working part of the population). The dependent part includes the population under 15 years old and people aged 65 and over. The productive part of a population therefore consists of those between 15 and 64 years who are in the workforce.

This ratio shows the pressure on the working population produced by the dependent part of the population.

The total dependency ratio of the population in South Africa is 51,9%. What does this value mean? It means that the working population (labour force) in South Africa must provide goods for itself and cover expenditure on children and aged persons. The value of more than 50% shows that less than half the population supports the rest.

[Adapted from: <http://xchmhlx.wordpress.com>]

SOURCE D



[Adapted from: <http://www.chrismadden.co.uk>]

SOURCE E

Life expectancy continues to rise as South Africa's population breaks 60 million

South Africa's population now stands at just over 62 million. This is the main finding of Stats SA's latest Mid-Year Population Estimates report, which was released in January 2023.

According to the report, life expectancy stands at 66 years, having increased from an estimated 52 years in 2005 and 61 years in 2011.

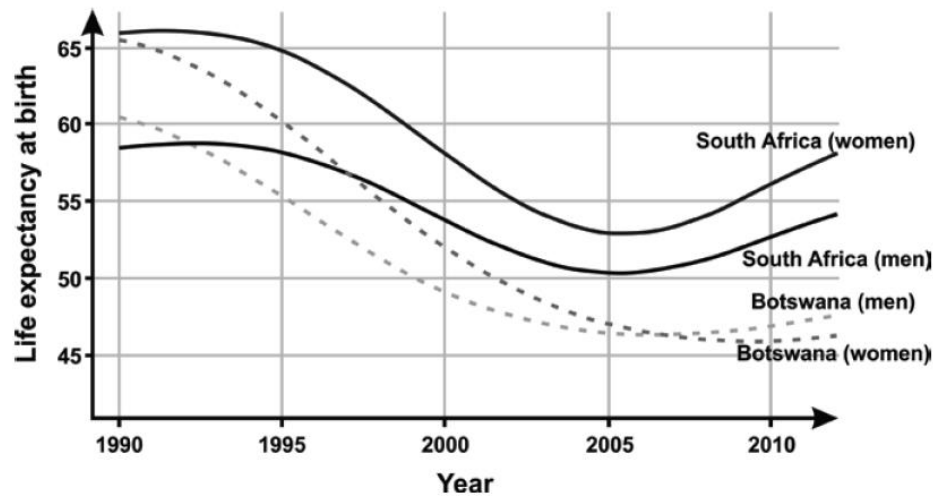
The rise in life expectancy can be attributed to two important trends:

- First, the number of AIDS-related deaths is estimated to have decreased from 250 000 deaths in 2015 (51% of all deaths) to 150 000 deaths in 2023 (31% of all deaths). This can be associated with the increased rollout of antiretroviral therapy.
- Second, the infant mortality rate has fallen from an estimated 34 infant deaths per 1 000 live births in 2014 to 23 infant deaths per 1 000 live births in 2023. The decline in the infant mortality rate points to an improvement in the general health and living standards of the population.



[Adapted from: <http://www.statssa.gov.za>]

Life expectancy in South Africa and Botswana with high HIV/AIDS prevalence, 1990–2012



[Adapted from: rpubs.com]

SOURCE F Sweden's Forced Sterilization Programme

For over 40 years, young socially marginalised working class women in Sweden faced the danger of forced sterilisation. This was carried out under laws intended to purify the Swedish race, prevent the mentally ill from reproducing and stamp out social activities classed as deviant. The last sterilisation took place in 1975.

Until the Sterilisation Act was finally repealed in 1975, 62 000 people, 90% of them women, were sterilised. Orphans were sterilised as a condition of their release from children's homes. Others were pinpointed on the basis of local neighbourhood gossip and personal grudges. Some were targeted because of their 'low intelligence', being of mixed race, or for physical defects.

The issue has become a national scandal, although similar revelations have emerged in other countries including Norway, Austria, Denmark, Finland, Belgium and the United States. Only Nazi Germany sterilised more people than Sweden.

[Adapted from: <http://www.wsws.org>]

SOURCE G Human Overpopulation



- Experts define overpopulation as the presence of excessive numbers of a species, which are then unable to be sustained by the space and resources available.
- The carrying capacity refers to the maximum number of organisms of a given species that an ecosystem can support.
- Environmental degradation, natural disasters, civil war, and forced resettlement may cause population increases in other locations.
- Internal displacement of large numbers of people, as well as undocumented migration across borders, may turn a region able to carry its original inhabitants into an overpopulated one.

The World Wildlife Fund has calculated that South Africans have an average ecological footprint of **2,8 hectares**. That is how much of our natural resources such as food, timber, land and energy we use to live our lives each year. But nature is only able to renew the supply of **1,8 hectares** per person, which means that we are running on overdraft.

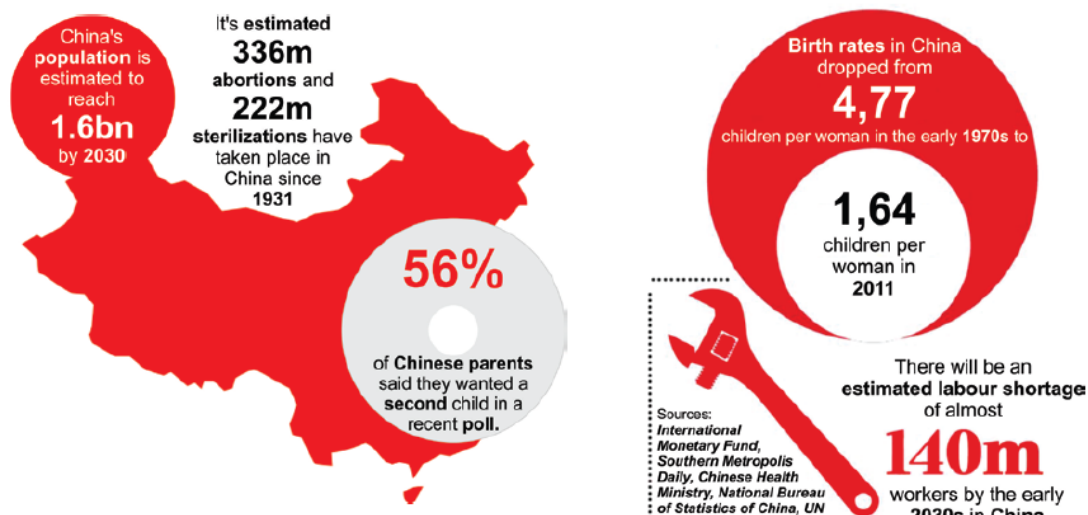
[Adapted from: <www.naturalstandard.com>]

SOURCE H

China Ends One-Child Policy, Allowing Families Two Children – October 2015

BEIJING – Driven by fears that an aging population could jeopardize China's economic growth, the Communist Party leadership ended its decades-old "one child" policy announcing that all married couples would be allowed to have two children.

The decision was a dramatic step away from a core Communist Party policy that was imposed in the late 1970s. This bred intense resentment over the brutal intrusions involved, including forced abortions and crippling fines, especially in the countryside.



[Image adapted from: <<http://nextbigfuture.com>>]

The efforts to limit family size also led to a skewed sex ratio of males to females. As traditional rural families favour boys over girls, they sometimes resorted to infanticide to ensure they had a son. Many young men now face the prospect of never finding a partner or starting a family of their own.

Yet while the decision surprised many experts and ordinary Chinese, some said it was unlikely to ignite either a baby boom or an economic one.

The initial public reaction to the party leaders' decision was restrained, and many citizens in Beijing who were asked whether they would grasp the chance to have two children expressed reluctance or outright indifference.

Some, however, were pleased.

Yang Qing, a clothing designer in Hangzhou, in eastern China, said that she wanted a second child, in addition to her 4-year-old son. "I always heard that people got fined for having more than one child, and they had to hide out and they will get caught," she said. "Everyone was scared."

"Really? Can you show me the news on your phone?" said Sun Bing, 34, the owner of a small technology store in Beijing, who had his 2-year-old son by his side. "This is a good thing, and I'm very supportive," he said. "I want to have a second kid in two years. But, of course, it's not cheap to raise children."

[Adapted from: <http://www.nytimes.com>]